Tezpur University: Spring Semester 2018: Formal Languages & Automata (CO205)
Test: III :: Marks: 25
Due Date: 19 April 2018, 5:00PM (*no extensions!*)

Please read the following instructions carefully:

- *Write your* ***name, roll number and course code*** *on top of the sheet.*
- *Write your answers in* ***neat and clear*** *handwriting!*
- ***Do not waste paper*** *in unnecessary embellishments/top sheets; write on both sides of the paper. It is only the content of your assignment sheet that carries marks!*

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1. Construct a CFG that recognizes the language L_1 where
   $L_1 = \{w | w \text{ starts and ends with the same symbol and } |w| \text{ is odd}\}$
   Convert the grammar to Chomsky Normal Form. [6+4]

2. Construct a Nondeterministic PDA that accepts the CFL L_2, where
   $L_2 = \{w | w \text{ has equal numbers of 0s and 1s}\}$
   Explain how the NPDA works, in your own words using (i) a string in the language and (ii) a string not in the language. [6+4]

3. Prove that the language L_3 is not context-free
   $L_3 = \{1^n 2^m 3^n 4^{n+m} | n, m \geq 1\}$ [5]